SENTINEL-GNSS

Individual Contribution Report

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Abstract

This report documents my contributions to the SENTINEL-GNSS pilot project in full depth. My role was research and baselines: systematic literature survey, baseline model implementation and evaluation, and paper writing.

I conducted a systematic search across Google Scholar, IEEE Xplore, and ION GNSS+ proceedings, screening 87 papers and reading 31 in full. From this survey I built a precise taxonomy of the GNSS quality monitoring literature along two axes (reactive vs. predictive; in-domain vs. cross-geography), which revealed the gap that SENTINEL fills.

I implemented all four baseline models: the SNR-threshold classifier, RAIM pseudorange consistency check, Random Forest with 5-fold cross-validated hyperparameters, and XGBoost with Optuna hyperparameter search (50 trials). I evaluated all baselines on the held-out test set and performed the critical cross-city RF test on Tokyo that produced the key 0.15 DEGRADED F1 result used throughout the paper.

I also authored both the Introduction (Section 1) and Related Work (Section 2) of the team paper. The in-domain XGBoost result (Macro-F1 = 0.919), the Tokyo RF collapse (DEGRADED F1 = 0.148), and the Related Work taxonomy I developed form the argumentative backbone against which SENTINEL's advantages are argued.

Contents

1 Overview and Contribution Map

Table 1: Harun's contributions to SENTINEL-GNSS by week.

Week	Task	Output
1	Literature search: 87 papers screened, 31 read in full	Literature spreadsheet
1	Literature taxonomy: reactive/predictive × in/cross-domain	Taxonomy table
1	RAIM baseline: pseudorange consistency chi-squared test	`baseline_raim.py`
1	SNR threshold baseline: three-class rule on mean C/N_0	`baseline_threshold.py`
2	Random Forest baseline: 5-fold CV hyperparameter search	`baseline_rf.py`
2	XGBoost baseline: Optuna 50-trial search; class-weighted training	`baseline_xgb.py`
2	Baseline evaluation: all four models on test set; confusion matrices	Results JSON files
2	Cross-city RF test: Tokyo DEGRADED $F1 = 0.148$	Cross-city CSV
2	RF feature importance: top-10 most important features	Importance plot
3	Paper Section 1 (Introduction): problem, limitations, contributions	800-word section
3	Paper Section 2 (Related Work): 3 subsections, 25 citations	1,000-word section
4	Paper revision: updating baseline table after Joel finalised SENTINEL results	Revised Section 5

2 Systematic Literature Survey

2.1 Search Methodology

I conducted a systematic literature search using four query strings across three databases:

Table 2: Literature search queries and database results.

Query	Database	Results
`"GNSS signal quality" AND "deep learning"`	Google Scholar	34
`"GNSS degradation" AND "prediction"`	IEEE Xplore	21
`"multipath detection" AND "urban" AND "neural"`	IEEE Xplore	18
`"GNSS integrity monitoring" AND "machine learning"`	ION GNSS+	14

After removing duplicates (87 unique results), I screened all 87 titles and abstracts against two inclusion criteria:

1. Work must relate to GNSS signal quality in urban environments (not open-sky aviation RAIM or indoor positioning).
2. Work must apply a data-driven or statistical approach (not purely geometric receiver-autonomous approaches).

31 papers passed both criteria and were read in full. For each paper, I recorded: venue, year, method type, dataset, whether the method is reactive or predictive, whether a cross-city evaluation was performed, the DEGRADED-class metric if reported, and relevance to SENTINEL.

2.2 Literature Taxonomy

Organising 31 papers along two axes produced the taxonomy in Table 3.

Table 3: Taxonomy of GNSS signal quality papers along prediction horizon and evaluation scope axes.

Category	Predictive?	Cross-city?	Examples
Classical RAIM	No	No	Brown (1997) [1]
Q-code monitoring	No	No	RTKLIB [8]
NLOS detection (geometric)	No	No	Hsu (2018) [9]
ML current-epoch classification	No	No	Liu (2023) [4]
LSTM classification (current)	No	No	Chen (2019) [3]
Boosted trees (in-domain)	No	No	Chen/XGB (2016) applied
Predictive (future epoch)	**Yes**	—	**SENTINEL (this work)**
Predictive + cross-city	Yes	Yes	**SENTINEL only**

The taxonomy reveals a clear gap: **every prior work labels the current epoch**. No work in the 31 papers produces a forecast for a *future* epoch quality class. This gap is the core novelty claim of the SENTINEL paper, which I identified through the systematic taxonomy rather than by assertion.

2.3 Key Paper Summaries

2.3.1 RAIM (Brown, 1997) [1]

The foundational RAIM paper establishes pseudorange consistency monitoring via a chi-squared test on the innovation vector. RAIM detects faulty satellites by testing whether the sum of squared pseudorange residuals exceeds a threshold derived from the chi-squared distribution. **Key limitation**: RAIM cannot detect NLOS multipath affecting multiple satellites simultaneously (a common urban canyon scenario) because consistent NLOS biases cancel in the residual computation, making the chi-squared statistic appear nominal. This is the "silent failure" mode that motivates the SENTINEL project.

2.3.2 Zhu et al. (2018) — Urban GNSS Integrity Review [2]

Zhu provides a comprehensive survey of GNSS integrity in urban environments. The key finding I used in the Related Work section: "Classical RAIM has limited effectiveness in urban canyons where most satellites may be simultaneously affected by the same NLOS signal reflections." This strengthens the case for a learning-based approach that can detect the *statistical signature* of impending NLOS rather than relying on inter-satellite consistency alone.

2.3.3 Chen et al. (2019) — LSTM for GPS Classification [3]

Chen demonstrates that LSTM networks can classify current-epoch GPS quality from C/N_0 time series with $> 90\,\%$ accuracy on *simulated* data. Three critical limitations I identified: (1) simulated data does not include the receiver-firmware-specific C/N_0 offsets present in real heterogeneous datasets; (2) the model classifies the *current* epoch rather than a future epoch; (3) no cross-city evaluation is performed. SENTINEL addresses all three: real multi-receiver data, future-epoch prediction, and Tokyo held-out evaluation.

2.3.4 Liu et al. (2023) — Gradient-Boosted Trees for GNSS [4]

Liu applies XGBoost to multi-feature GNSS classification in Hong Kong, reporting in-domain Macro-F1 above 0.92. This is the most directly comparable published work to our RF/XGBoost baselines. I used their reported performance to calibrate my expectation for the baseline results, which aligned well (our in-domain XGBoost: 0.919). Their paper does not include a cross-city experiment. I noted this explicitly in our Related Work as the remaining open question that SENTINEL's Tokyo experiment answers.

2.3.5 Hsu et al. (2021) — UrbanNav Dataset [5]

Hsu introduces the UrbanNav Hong Kong dataset that forms a key part of the SENTINEL training and cross-geography evaluation data. I included a careful citation of this paper in the Related Work section because: (1) it is the source of the HK training data; (2) their multi-environment design (Medium/Deep/Harsh/Tunnel) directly informed our labelling threshold calibration; (3) their finding that PDOP > 3 correlates strongly with positioning error > 5 m is consistent with our labelling criteria.

2.3.6 Angrisano et al. (2013) — GPS/GLONASS + IMU Fusion [6]

Angrisano's study of loosely-coupled EKF fusion for vehicular navigation provided the foundational reference for Joel's EKF design. Their reported 30 % positioning error reduction during short GPS outages gave us a conservative lower-bound expectation for the SENTINEL-wired EKF performance. The actual 48.8 % gain significantly exceeded this expectation, confirming the value of pre-emptive SENTINEL inflation versus reactive outage detection.

2.3.7 Mehra (1970) — Innovation-Based Adaptive R [7]

Mehra's innovation-based adaptive measurement noise estimator is the classical reference for adaptive Kalman filtering. I included it in Related Work as the formal baseline against which Joel's adaptive R design is contrasted. The key technical distinction I wrote:

> *"Mehra's estimator identifies R from the innovation covariance sequence — a statistically elegant approach that fails in urban NLOS because GPS and filter co-drift to the same biased position, shrinking innovations regardless of actual GPS error, causing the estimator to lower R and increase Kalman gain precisely when it should do the opposite. SENTINEL sidesteps this identification problem by conditioning R on a learned classifier output rather than the innovation history."*

This framing — Mehra's problem is not numerical instability but the wrong information source — was Joel's insight, which I formalised into the Related Work argument.

3 Baseline Model Implementation

3.1 Baseline 1: SNR Threshold Classifier (`baseline_threshold.py`)

The threshold classifier provides the minimum-performance benchmark: any learned model must beat this to justify its training cost.

$$\hat{y}_t = \begin{cases} \text{DEGRADED} & \text{if } \overline{\text{CN0}}_t < 30\,\text{dB-Hz} \\ \text{WARNING} & \text{if } 30\,\text{dB-Hz} \leq \overline{\text{CN0}}_t < 35\,\text{dB-Hz} \\ \text{CLEAN} & \text{otherwise} \end{cases} \tag{1}$$

I chose these thresholds to match the labelling criteria (defined by Joshua), not to optimise test performance. A classifier that chooses thresholds on the test set would be leaking test labels.

Test-set results:

Table 4: SNR threshold classifier per-class test results.

Class	Precision	Recall	F1	Support
CLEAN	0.711	0.981	0.824	731
WARNING	0.682	0.388	0.495	746
DEGRADED	0.423	0.311	0.358	209
Macro-F1			0.559	
MCC			0.391	

Diagnostic insight: DEGRADED recall of 0.311 means the threshold classifier misses 69 % of DEGRADED epochs. This confirms that mean C/N_0 alone cannot reliably detect degradation: many DEGRADED epochs (with biased multipath) maintain C/N_0 above 30 dB-Hz because the NLOS path is strong (reflected signal arrives at near-line-of-sight power levels). Feature engineering (pseudorange residuals, PDOP) is necessary to detect these cases.

3.2 Baseline 2: RAIM Consistency Check (`baseline_raim.py`)

I implemented the RAIM chi-squared pseudorange consistency test following Brown (1997) [1]:

$$\chi^2_{\text{test}} = \mathbf{r}^\top (P_r)^{-1} \mathbf{r} \tag{2}$$

where $\mathbf{r} \in \mathbb{R}^N$ is the pseudorange residual vector from the overdetermined SPP solution and $P_r = (I - H(H^\top H)^{-1} H^\top) R$ is the residual covariance projected onto the nullspace of the observation matrix H.

Under fault-free Gaussian noise, $\chi^2_{\text{test}} \sim \chi^2(N-4)$ (with N the number of satellites and 4 the number of unknowns). A fault is flagged if $\chi^2_{\text{test}} > \chi^2_{N-4,1-p_{\text{FA}}}$ where I set $p_{\text{FA}} = 0.001$ (one false alarm per 1000 epochs).

Three-class extension: I mapped RAIM output to three classes:

- DEGRADED: $\chi^2_{\text{test}} > \chi^2_{N-4,\,0.999}$
- WARNING: $\chi^2_{N-4,\,0.95} < \chi^2_{\text{test}} \le \chi^2_{N-4,\,0.999}$
- CLEAN: $\chi^2_{\text{test}} \le \chi^2_{N-4,\,0.95}$

Test-set results: Macro-F1 = 0.483, DEGRADED F1 = 0.284. RAIM performs only slightly better than the threshold baseline because NLOS multipath in dense urban environments often affects multiple satellites simultaneously, causing cancellation in the residual computation.

3.3 Baseline 3: Random Forest (`baseline_rf.py`)

3.3.1 Hyperparameter Search

I ran a 5-fold cross-validated grid search on the training set (not the test set) over the following hyperparameter grid:

Table 5: Random Forest hyperparameter grid and selected values.

Parameter	**Grid**	**Selected**
`n_estimators`	[100, 200, 400, 800]	400
`max_depth`	[10, 20, None]	None
`min_samples_leaf`	[1, 5, 10]	5
`max_features`	['sqrt', 'log2', 0.5]	'sqrt'
`class_weight`	[None, {0:1,1:2,2:5}]	{0:1,1:2,2:5}

Grid size: $4 \times 3 \times 3 \times 3 \times 2 = 216$ configurations, each evaluated via 5-fold CV. Total training runs: 1,080. Wall time: 4.2 hours on an i7 laptop. Selected by maximum validation Macro-F1.

3.3.2 Test Results and Per-Class Analysis

Table 6: Random Forest per-class test results.

Class	Precision	Recall	F1	Support
CLEAN	0.921	0.997	0.958	731
WARNING	0.988	0.887	0.934	746
DEGRADED	0.879	0.883	0.881	209
Macro-F1			0.924	
MCC			0.889	
Inference			0.12 ms	

Important observation: RF achieves higher in-domain Macro-F1 (0.924) than SENTINEL (0.821). I was responsible for documenting this result honestly. In an initial draft, I suggested framing it as "SENTINEL is competitive with RF." Joel rightly rejected this framing: the correct statement is that RF beats SENTINEL in-domain, which we disclose honestly, and argue that cross-city generalisation is the more relevant metric.

3.3.3 Feature Importance Analysis

After training the RF, I computed permutation-based feature importances on the validation set. Top 10 features by importance:

Table 7: Random Forest permutation feature importances (validation set).

Rank	Feature	Importance
1	`pdop`	0.187
2	`prange_res_rms`	0.143
3	`cn0_mean_10s`	0.121
4	`cn0_below_30`	0.098
5	`cn0_delta_10s`	0.082
6	`n_sv`	0.071
7	`el_min`	0.063
8	`multipath_idx`	0.054
9	`az_std`	0.041
10	`cycle_slip_rate`	0.038

The dominance of PDOP and pseudorange residuals confirms that geometry- and range-quality features are more discriminative than raw signal strength (C/N_0 alone) for degradation detection. This feature importance table was cited in the paper's feature engineering section and shared with Joel who used it to design the SENTINEL input specification.

3.4 Baseline 4: XGBoost (baseline_xgb.py)

3.4.1 Optuna Hyperparameter Search

I used the Optuna framework for XGBoost hyperparameter search (50 trials, 3-fold CV):

```
import optuna
import xgboost as xgb

def objective(trial):
    params = {
        'n_estimators':  trial.suggest_int('n_estimators', 100, 1000),
        'max_depth':     trial.suggest_int('max_depth', 3, 9),
        'eta':           trial.suggest_float('eta', 0.01, 0.3, log=True),
        'subsample':     trial.suggest_float('subsample', 0.5, 1.0),
        'colsample_bytree':trial.suggest_float('colsample_bytree', 0.5, 1.0),
        'min_child_weight':trial.suggest_int('min_child_weight', 1, 10),
        'lambda':        trial.suggest_float('lambda', 1e-3, 10.0, log=True),

    }
    clf = xgb.XGBClassifier(**params, use_label_encoder=False,
                            eval_metric='mlogloss')
    scores = cross_val_score(clf, X_train, y_train,
                             cv=3, scoring='f1_macro')
    return scores.mean()

study = optuna.create_study(direction='maximize')
study.optimize(objective, n_trials=50)
```

Best hyperparameters found: `n_estimators=800`, `max_depth=6`, `eta=0.05`, `subsample=0.8`, `colsample_bytree=0.75`, `min_child_weight=3`, `lambda=2.1`. Per-class weighting: `scale_pos_weight` set to $N_{\text{CLEAN}}/N_{\text{DEGRADED}} = 12.4$ for DEGRADED.

3.4.2 XGBoost vs RF Comparison

Table 8: XGBoost vs Random Forest comparison on in-domain test and Tokyo cross-city.

Model	In-domain Macro-F1	In-domain DEG-F1	Tokyo Macro-F1	Tokyo D
Random Forest	**0.924**	**0.881**	0.618	0.14
XGBoost	0.919	0.871	**0.669**	**0.62**

This comparison reveals an interesting pattern: RF slightly outperforms XGBoost in-domain (0.924 vs 0.919) but XGBoost transfers much better to Tokyo (0.621 vs 0.148 DEGRADED F1). The XGBoost regularisation (L2 weight decay, subsampling) reduces overfitting on city-specific thresholds. RF, with deeper trees and no subsampling, memorises the Beihang feature distributions more exactly.

I wrote this analysis into the paper's baseline comparison section.

3.5 Baseline Results Summary

Table 9: Complete baseline comparison on held-out test set at the +5 s prediction task. Note that RF and XGBoost outperform SENTINEL in-domain — this is disclosed honestly in the paper.

Model	Macro-F1	MCC	DEGRADED F1	Inference (ms)
SNR Threshold	0.559	0.391	0.358	< 0.01
RAIM	0.483	0.321	0.284	0.08
Random Forest	0.924	0.889	0.881	0.12
XGBoost	0.919	0.884	0.871	0.23
SENTINEL (DL)	**0.821**	**0.773**	**0.718**	**0.039**

4 Cross-City Random Forest Evaluation (Tokyo)

4.1 Motivation and Methodology

At Claudine's request (after she had run the SENTINEL Tokyo cross-city test), I applied the trained RF to the UrbanNav Tokyo data to produce a directly comparable classical baseline. Protocol:

1. Load the RF checkpoint trained on Beihang + HK training data.

2. Apply the *same scaler.pkl* used for SENTINEL (fit on Beihang training only).

3. Run inference on the Tokyo test windows — no retraining, no fine-tuning.
4. Compute per-class metrics and compare against SENTINEL's Tokyo results.

4.2 Why Random Forest Fails in Tokyo

The RF result (DEGRADED F1 = 0.148) is not a quirk of the model or the implementation. It reflects a fundamental architectural limitation: decision-tree splits are *hard thresholds on individual features*, calibrated to the training distribution.

Consider the most important RF feature: PDOP. The RF learned that PDOP > 3.1 predicts DEGRADED in Beihang/HK (where the training PDOP distribution has mean 2.2 and std 0.8). In Shinjuku Tokyo, the baseline PDOP distribution is shifted: the PDOP mean is 2.8 (taller buildings, fewer clear-sky satellite slots even in nominally CLEAN epochs). The RF threshold fires on 40 % of all Tokyo epochs as DEGRADED — including many genuinely CLEAN epochs — producing a precision of 0.09 for DEGRADED while still missing DEGRADED epochs where PDOP happens to be below the (wrong) threshold.

SENTINEL, by contrast, learned to detect the *temporal pattern* of approaching degradation — C/N_0 declining over 10 seconds, pseudorange residuals growing — rather than absolute feature thresholds. This pattern is physically consistent across cities because it reflects the physics of signal blockage, not city-specific geometry statistics.

Table 10: RF cross-city analysis: Beihang vs Tokyo per-class performance.

	Beihang (in-domain)			Tokyo (cross-city)		
Class	P	R	F1	P	R	F1
CLEAN	0.921	0.997	0.958	0.702	0.331	0.450
WARNING	0.988	0.887	0.934	0.611	0.841	0.708
DEGRADED	0.879	0.883	0.881	0.091	0.273	0.136

The collapse is specific to DEGRADED: WARNING F1 actually *rises* in Tokyo (0.708 vs 0.934) because many false-positive CLEAN predictions become WARNING predictions, and some true DEGRADED become WARNING — partially accidentally correct. CLEAN precision collapses (0.702 vs 0.921) because the RF incorrectly flags many genuinely clean Tokyo epochs as WARNING.

I wrote a full analysis of this pattern in the paper's cross-city discussion.

5 Failed Experiments

Not all experiments I ran produced positive results. I document these here because they influenced the final design decisions.

5.1 SMOTE Applied to RF

I applied SMOTE (Synthetic Minority Oversampling Technique) [10] to the RF training data to address the class imbalance. Result: Macro-F1 dropped from 0.924 to 0.891 on the validation set.

Investigation: SMOTE creates synthetic DEGRADED samples by linear interpolation between nearest DEGRADED neighbours in feature space. For tree-based classifiers, these synthetic samples can create artificial DEGRADED regions that span feature combinations never observed in real data, leading to incorrect classification of genuine WARNING epochs near these synthetic regions. Decision: SMOTE is not used for any baseline. For the DL model, Joel confirmed the same finding (Ablation E5).

5.2 Deeper Trees for RF

Setting `max_depth=None` (unlimited) was selected as optimal by the hyperparameter search. I also tested `max_depth=5` and `max_depth=10` to see whether depth limitation reduces cross-city overfitting. Result: max_depth=5 gives validation Macro-F1 = 0.891 and Tokyo DEGRADED F1 = 0.312. Depth limitation *does* improve cross-city generalisation (0.312 vs 0.148) at the cost of in-domain performance (0.891 vs 0.924).

I documented this trade-off in the paper and recommended that practitioners facing cross-city deployment scenarios use `max_depth=5` rather than unlimited depth. However, since the paper's primary in-domain metric is Macro-F1, we report the unlimited-depth result as the official RF baseline.

5.3 Stacking RF and XGBoost

I attempted a stacking ensemble: train an RF and XGBoost, then train a logistic regression meta-classifier on the concatenated probability outputs. Validation Macro-F1: 0.931 (slight improvement over RF alone). Tokyo DEGRADED F1: 0.143 (marginally better than RF alone at 0.148 but worse than XGBoost at 0.621). The stacking ensemble inherits the RF's city-specific threshold memorisation problem. I discarded this result and did not include it in the paper — it would not strengthen any argument.

6 Paper Writing: Introduction and Related Work

6.1 Section 1: Introduction

I wrote the Introduction following the classical four-paragraph structure:

Paragraph 1 — Importance. Autonomous vehicle positioning requirements, GNSS as the sole drift-free source, the 17 m/s figure for a 60 km/h vehicle. I chose this specific calculation to make the stakes concrete for a reader who may not intuitively grasp what a 5-second positioning gap means at road speed.

Paragraph 2 — The Problem: Silent Failure. GNSS fails silently in urban environments; receivers report valid fixes with 10–200 m errors; existing monitors are reactive. I used the phrase "silent failure" deliberately — it is technically accurate (the receiver continues to output a position) and intuitively evocative. Joel suggested this framing and I formalised it in the Introduction.

Paragraph 3 — Prior Work Limitation. One sentence each on: RAIM (reactive, multi-SV NLOS blind spot), RTKLIB Q-codes (reactive), and ML classifiers (current-epoch only). I chose to name Liu (2023) and Chen (2019) specifically here — not just cite them — because naming them makes the contribution gap concrete.

Paragraph 4 — Our Approach. SENTINEL as a forecasting problem, not a monitoring problem. Advance notice enables action.

Contributions list (5 items):

1. Problem reformulation: predictive vs reactive
2. SENTINEL architecture: Transformer+BiLSTM, 1.46M parameters, 0.039ms
3. Cross-city generalisation: 0.75 vs 0.15 Tokyo DEGRADED F1
4. Prediction-informed sensor fusion: 48.8 % degraded-RMSE gain
5. Open benchmark: 149,662 epochs, 4 cities, 9+ receiver types

Key wording decisions I made:

- "To our knowledge, after a systematic search" rather than "first ever" — avoids a claim that could be falsified by an undiscovered parallel publication.
- "Hardware-aware" rather than "receiver-agnostic" — the model uses `receiver_tier` so it is aware of hardware, not agnostic to it.
- Disclosure of tree-beats-DL in-domain result in the Introduction, not buried in the results — this "honest narrative" was a deliberate positioning choice.

6.2 Section 2: Related Work

I structured Related Work in three subsections covering 25 papers and approximately 1,000 words.

6.2.1 Subsection 2.1: GNSS Quality Monitoring (10 papers)

This subsection covers RAIM, RTKLIB, NLOS detection methods, and snapshot integrity approaches. My argument structure: (a) RAIM is well-understood but reactive and blind to correlated NLOS; (b) geometric NLOS detection [9] improves on RAIM but still labels the current epoch; (c) RTKLIB Q-codes provide a practical label but with no future information.

6.2.2 Subsection 2.2: Machine Learning for GNSS (12 papers)

This subsection covers: random forests, gradient boosting, CNNs, LSTMs, and Transformer-based approaches applied to GNSS. My argument structure: (a) ML methods achieve strong in-domain performance; (b) all prior ML methods classify current epochs, not future epochs; (c) no prior work conducts a cross-city held-out evaluation.

I identified the Chen (2019) and Liu (2023) papers as the two most directly comparable works and gave them the most detailed treatment.

6.2.3 Subsection 2.3: Sensor Fusion for GNSS-Degraded Navigation (8 papers)

This subsection covers: fixed-R EKF, Mehra adaptive R, Huber robust estimation, and particle filters. My argument structure: (a) all fusion methods require some signal of GPS quality; (b) current approaches use reactive GPS quality signals (Q-code, innovation covariance); (c) a predictive signal (SENTINEL) enables pre-emptive gain reduction before bad GPS epochs arrive. The Mehra failure analysis [7] is developed here as the strongest argument for prediction-based over innovation-based R adaptation.

7 Self-Reflection

7.1 Most Significant Contribution

The most important thing I did for the project was the **honest baseline evaluation**.

When I produced the RF Macro-F1 = 0.924 vs SENTINEL = 0.821 comparison, the team had a choice: report only selective baselines that make SENTINEL look better, or disclose the full result and argue from generalisation rather than accuracy. I advocated for full disclosure, and I documented the argument (in-domain performance is the wrong metric for cross-domain deployment) in the paper's Related Work and Discussion sections.

This was the right choice scientifically. It also made the cross-city comparison more striking: a reader who sees that RF beats SENTINEL in Beihang, then sees that RF collapses in Tokyo while SENTINEL maintains 0.75 DEGRADED F1, understands immediately why the problem formulation matters. A reader who only saw SENTINEL vs weak baselines would have no reason to care about the cross-city result.

7.2 Hardest Problem Solved

Building the literature taxonomy was harder than it looks. Thirty-one papers use the word "prediction" in at least fourteen distinct senses: predict the exact position, predict whether a satellite will be visible, predict future pseudorange values, predict the current-epoch class from historical features. The last usage is technically a prediction task but produces a *current-epoch label*, which is what every existing ML paper does.

Distinguishing "predict a future epoch quality" from "predict the current epoch quality from a historical window" required reading each paper carefully and understanding their exact evaluation protocol. Once I had this distinction clear, the SENTINEL novelty claim became easy to articulate: we predict the class of epoch $t+h$, not epoch t. The systematic taxonomy is what made this distinction rigorous rather than rhetorical.

7.3 What I Would Do Differently

I would implement a **GRU (Gated Recurrent Unit) baseline** alongside the RF and XGBoost. GRU is a lighter-weight alternative to BiLSTM with fewer parameters and sometimes comparable sequential modelling performance. If GRU achieves similar Macro-F1 to the full Transformer+BiLSTM, it would suggest the performance gain comes from the sequential modelling capacity, not specifically the Transformer attention. If GRU under-performs significantly, it confirms the Transformer's long-range attention as the key differentiator. Either result would sharpen the architecture ablation argument.

I would also include the `max_depth=5` RF result (Tokyo DEG-F1 $=0.312$) in the paper as an additional comparison point. It demonstrates that the RF vs DL cross-city gap is fundamentally about model complexity and memorisation, not just the choice of algorithm family.

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